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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-23

Programme: B.Tech /MBA Tech (Data Science) Year: II Semester: III

Subject: Managing Uncertainty

Date: 21 November 2022

Marks: 100
Time: 10.00 am - 1.00 pm
Durations: 3 (Hrs)
No. of Pages: 05+1=6

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

<table>
<tr><td>Q1</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>CO- 2;
SO-1 ;
BL-1</td><td>A</td><td>Suppose that samples of polythene bags from two manufacturers A and B are tested by a buyer for bursting pressure, giving the following results
<table>
<tr><td></td><td colspan="2">Number of Bags</td></tr>
<tr><td>Bursting Pressure</td><td>A</td><td>B</td></tr>
<tr><td>5 -10</td><td>2</td><td>9</td></tr>
<tr><td>10 -15</td><td>9</td><td>11</td></tr>
<tr><td>15 – 20</td><td>29</td><td>18</td></tr>
<tr><td>20 – 25</td><td>54</td><td>32</td></tr>
<tr><td>25 – 30</td><td>11</td><td>27</td></tr>
<tr><td>30 – 35</td><td>5</td><td>13</td></tr>
</table></td><td>2

3</td></tr>
</table>

<table>
<tr><td></td><td></td><td>I. Which set of bags has the on an average highest bursting pressure?

II. Which has more uniform pressure? If prices are the same, which manufacture's bag would be preferred by the buyer? Why?</td><td></td></tr>
<tr><td>CO-2 ;
SO-2 ;
BL-2</td><td>B</td><td>In rolling two dice, what is the probability of rolling

I. A total of 7 on the first roll, followed by a total of 11 on the second roll?

II. A total of 21 on the first two rolls combined?

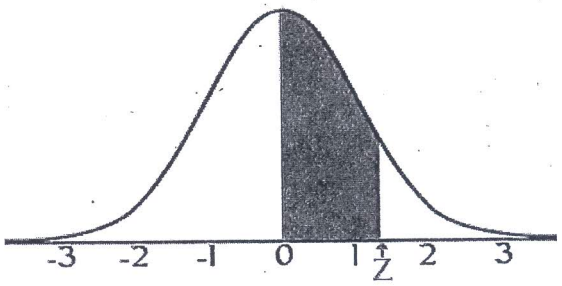
III. A total of 6 on the first three rolls combined?</td><td>2
2

1</td></tr>
<tr><td>CO-1 ;
SO-1 ;
BL-2</td><td>C</td><td>Briefly describe Frequency Polygons.</td><td>5</td></tr>
<tr><td>CO- 3;
SO-2 ;
BL-1</td><td>D</td><td>In an attempt to estimate potential future demand, the National Motor Company did a study asking married couples how many cars the average energy-minded family should own in 1998. For each couple, National averaged the husband's and wife's response to get the overall couple response. The answers were then tabulated.
<table><tr><td>Number of cars</td><td>0</td><td>0.5</td><td>1.0</td><td>1.5</td><td>2.0</td><td>2.5</td></tr><tr><td>Frequency</td><td>2</td><td>14</td><td>23</td><td>7</td><td>4</td><td>2</td></tr></table>Calculate the variance and the standard deviation.</td><td>5</td></tr>
<tr><td>Q2</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q2
CO-2;
SO-1;
BL-1</td><td>A</td><td>Hinton Press hypothesizes that the average life of its largest web press is 14,500 hours. They know that the standard deviation of press life is 2,100 hours. From a sample of 25 presses, the company finds a sample mean of 13,000 hours. At a 0.01 significance level, should the company conclude that the average life of the presses is less than the hypothesized 14,500 hours?</td><td>10</td></tr>
</table>

<table>
<tr><td>Q2
CO-2;
SO-1;
BL-2</td><td>B</td><td>A study of the age of 100 persons grouped into intervals 20-22, 22-24, 24-26,…., revealed the mean age and standard deviation to be 32.02 and 13.18 respectively. While checking it was discovered that the observation 57 was misread as 27. Calculate the correct mean age and standard deviation.</td><td>10</td></tr>
<tr><td>Q3</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q3
CO-3;
SO-2; BL-3</td><td>A</td><td>Calculate the standard deviation and coefficient of variation from the following data.
<table><tr><th>Marks</th><th>No of students</th><th>Marks</th><th>No of students</th></tr><tr><td>Up to 10</td><td>12</td><td>Up to 50</td><td>157</td></tr><tr><td>Up to 20</td><td>30</td><td>Up to 60</td><td>202</td></tr><tr><td>Up to 30</td><td>65</td><td>Up to 70</td><td>222</td></tr><tr><td>Up to 40</td><td>107</td><td>Up to 80</td><td>230</td></tr></table></td><td>10</td></tr>
<tr><td>Q3
CO-2;
SO-1; BL-2</td><td>B</td><td>Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT? How high must an individual score on the GMAT in order to score in the highest 5%?</td><td>10</td></tr>
<tr><td>Q4</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q4
CO-;2
SO-2;
BL-1</td><td>A</td><td>A company wants to improve sales. Past sales data indicate that the average sale was $100 per transaction. After training the sales force, recent sales data (taken from a sample of 25 salesmen) indicates an average sale of $130, with a standard deviation of $15. Did the training work? Test your hypothesis at a 5% significance level.</td><td>10</td></tr>
<tr><td>Q4</td><td></td><td>Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be
I. 0 jobs;</td><td>10</td></tr>
</table>

CO-;2 SO-2; BL-1	B	II. exactly 2 jobs; III. at most 3 arrivals. IV. What is the maximum jobs that should arrive one minute with 90 % certainty? V. more than 3 Jobs	
Q5		Answer briefly:	20
Q5 CO-2; SO-2; BL-3	A	Two dice are rolled. A = 'sum of two dice equals 3' B = 'sum of two dice equals 7' C = 'at least one of the dice shows a 1' I. What is P(A|C)? II. What is P(B|C)? III. Are A and C independent? What about B and C?	3 3 4
Q5 CO-2; SO-2; BL-2	B	In its third year, attendance in the Liberty Football League averaged 16,050 fans per game,and had a standard deviation of 2,500. I. According to these data, what is the probability that the number of fans at any given game was greater than 20,000? II. Fewer than 10,000? III. Between 14,000 and 17,500?	10
Q6		Answer briefly:	20
Q6 CO-2; SO-2; BL-3	A	Hunter says she is itchy. There is a test for Allergy to Cats, but this test is not always right: • For people that really do have the allergy, the test says "Yes" 80% of the time • For people that do not have the allergy, the test says "Yes" 10% of the time ("false positive") If 1% of the population have the allergy, and Hunter's test says "Yes", what are the chances that Hunter really has the allergy?	10
Q6		You are planning a picnic today, but the morning is cloudy • Oh no! 50% of all rainy days start off cloudy!	10

CO-2; SO-2; BL-2	B	• But cloudy mornings are common (about 40% of days start cloudy) • And this is usually a dry month (only 3 of 30 days tend to be rainy, or 10%) What is the chance of rain during the day?	
Q7		Answer briefly:	20
Q7 CO-2; SO-2; BL-2	A	I. Explain continuous random variables. Also, give and explain their PDF. II. State and explain joint probability distribution for continuous random variables.	6 4
Q7 CO-2; SO-2; BL-1	B	In the past 20 years, on average, only 3 percent of all checks written to the American Heart Association have bounced. This month, the A.H.A. received 200 checks. What is the prob- ability that I. Exactly 10 of these checks bounced? II. Exactly 5 of these checks bounced?	10



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998